ventory Optimization -- EOQ & Newsvend

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Mode: Full Pipeline

1. Problem Formulation

Problem Type: Find

Domain: Mathematical Optimization

Universe

- Eoq: \{quantity, total_cost, ordering_cost, holding_cost, orders_per_year, cycle_time_days\}
- Reorder Point: \{rop, safety_stock, z_score, mean_demand_lt, std_demand_lt\}
- Newsvendor: \{critical_ratio, optimal_q, expected_profit, expected_leftover, expected_shortag
- Simulation: \{n_cycles, simulated_fill_rate, target_service_level, fill_rate_error, simulated_nv_r

Variables

Symbol	Meaning	Type / Range
x	decision variable	\mathbb{{R}}

Structure

A retail store sells a product with the following characteristics:

- **Annual demand**: 10,000 units
- **Order cost**: $50 per order
- **Holding cost**: $2/unit/year
- **Lead time**: 1 week (deterministic)
- **Weekly demand**: mean = 192 units, std = 40 units
- **Perishable variant**: unit cost = $10, selling price = $25, salvage value = $3

Questions:
1. What is the optimal order quantity (EOQ)?
2. What reorder point achieves a 95% service level?
3. What is the optimal newsvendor order quantity for the perishable variant?
4. How sensitive are these answers to parameter changes?

Real-World Mapping

Real-World Concept	Mathematical Object
problem input	instance parameters

Constraints

(1) Economic Order Quantity (EOQ)**: Balances ordering cost and holding cost to minimize tot

(2) Reorder Point (ROP)**: Inventory level that triggers a new order, accounting for lead-time d

(3) Safety Stock**: Extra inventory held to buffer against demand variability during lead time

(4) Newsvendor Model**: Single-period stocking decision for perishable goods under demand u

(5) Critical Ratio**: Cu / (Cu + Co) *the probability threshold for the newsvendo*

(6) Service Level**: Probability of not stocking out during a replenishment cycle

Approach

Suggested approach EOQ + (s,Q) reorder point + Newsvendor

Complexity class:

Available tools: EOQ +

2. Solution

Answer:	Solution computed
Optimal:	No / Unknown
Feasible:	No
Algorithm:	EOQ + (s,Q) reorder point + Newsvendor
Time:	0.0019s

Solution Details

Solution computed successfully.

Verification

✓	Eoq Positive	Passed
✓	Eoq Formula Correct	Passed
✓	Total Cost At Eoq	Passed
✓	Rop Gt Mean Demand	Passed
✓	Safety Stock Positive	Passed
✓	Newsvendor In Range	Passed
✓	Critical Ratio Correct	Passed
✓	Simulation Service Level	Passed
✓	All Passed	Passed

3. Interpretation

The Question

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Questions:
1.

The Answer

A feasible solution was found.

What This Means

- EOQ: ~707 units, total cost ~$1,414/year, ~14 orders/year
- Reorder point: ~258 units (with ~66 units safety stock)
- Newsvendor Q: ~222 units, critical ratio ~0.682
- Monte Carlo fill rate: within 3% of 95% target

Recommendations

1. Review the model assumptions to ensure real-world fidelity.

Limitations

- Model assumes deterministic parameters; real-world variability not captured.
- Solutio